Game Theory

Lecture 1

Reference: *An Introduction to Game Theory*

Martin J. Osborne, Oxford University Press, 2003.

Structure of the Lecture

- Definition of Games in Strategic Form.

- Solution Concepts

  Nash Equilibrium, Dominance and Rationalizability.
- Strategic games with perfect information

- Applications

  Cournot Oligopoly, Bertrand Duopoly, Stackelberg Model from Mix strategy, Downsian Electoral Competition, Vickrey Second Price Auction.

Game Theory

- Game theory was developed by John Von Neumann and Oscar Morgenstern in 1944 -
  - Economists!
- One of the fundamental principles of game theory, the idea of equilibrium strategies was developed by John F. Nash, Jr. (A Beautiful Mind)
- Game theory is a way of looking at a whole range of human behaviors as a game.

Components of a Game

- Games have the following characteristics:
  - Players
  - Rules
  - Payoffs
  - Strategies

Types of Games

- We classify games into several types.
  - By the number of players:
  - By the Rules:
  - By the Payoff Structure:

Games as Defined by the Number of Players:

- 1-person (or game against nature)
- 2-person
- n-person(3-person up)

Games as Defined by the Rules:

- These determine the number of options/alternatives in the play of the game.
- The payoff matrix has a structure (independentof value) that is a function of the rules of the game.
- Thus many games have a 2x2 structure due to 2 alternatives for each player.

Games as Defined by the Payoff Structure

- Zero–sum
- Non–zero sum
- (and occasionally Constant sum)
  - Examples:
    - Zero–sum

      Classic games: Chess, checkers, tennis, poker.

      Political Games: Elections, War

    Non–zero sum

      Classic games: Football (?), D&D, Video games

      Political Games: Policy Process

Strategies

- We also classify the strategies that we employ:
- It is natural to suppose that one player will attempt to anticipate what the other player will do. Hence
  - Minimax - to minimize the maximum loss - a defensive strategy
  - Maximin - to maximize the minimum gain - an offensive strategy.

Again: What is Game Theory?

- Game Theory is the formal study of strategic interactions.
- A strategic interaction involves two or more agents. They maximize their payoffs and are aware that their opponents maximize payoffs.
- Applications range from economics to politics, to biology and computer science.

Games in Strategic Form

A game in strategic form is

- a set of players: {1, 2, …, I}
- for each player *i,* a set S_i of strategies s_i
- for each player *i,* preferences over the set of strategy profiles $S=\{(s_1, \ldots, s_I)\}$, represented by

  $u : S \rightarrow R^I$

  (a strategy profile includes one strategy for each player).

Games in Strategic Form: examples

- Wide range of situations may be modelled as strategic games. For example:

1. the players may be firms, the actions prices, and the preferences a reflection of the firms' profits.
2. the players may be candidates for political office, the actions campaign expenditures, and the preferences a reflection of the candidates' probabilities of winning.
3. Similarly the players may be animals fighting over some prey, the actions concession times, and the preferences a reflection of whether an animal wins or loses

Games in Strategic Form: continued…

The theory of rational choice

The theory of rational choice is a component of many models in game theory.

- It states that "**a decision-maker chooses the best action according to his/her preferences, among all the actions available to him/her**".
- No qualitative restriction is placed on the decision-maker's preferences; her "rationality" lies in the consistency of his/her decisions when faced with different sets of available actions, not in the nature of his/her likes and dislikes.
- How can we describe a decision-maker's preferences?
- we can "represent" the preferences by a ***payoff function***, which associates a number with each action in such a way that actions with higher numbers are preferred.
- The payoff function ***u*** represents a decision-maker's preferences if, for any actions ***a*** in *A* and ***b*** in *A*,

$$u(a) > u(b) \text{ if and only if the decision-maker prefers } a \text{ to } b.$$

(A better name than payoff function might be "preference indicator function"; in economic theory a payoff function that represents a consumer's preferences is often referred to as a "utility function".)

Solution Concepts

- A solution concept is a mathematical rule to find the solution of a game.

- It allows the modeler to formulate a prediction on the play of the interaction she modeled as a game.

- we will study 3 solution concepts:
  - Nash Equilibrium
  - Dominance
  - Rationalizability

Prisoner's Dilemma

- Two prisoners are separately interviewed. By accusing the other suspect, one's prison term is reduced. But if they both stayed quiet, they would not be incarcerated.

- *Players:* The two suspects.

- *Strategies:* Each player's set of strategy is *{Quiet*, *Fink}*.

- *Preferences:* Suspect 1's ordering of the strategy
- profiles, from best to worst is (*F*, *Q*), (*Q*, *Q*), (*F, F)*, (*Q*, *F*).

- Payoff functions that represent the suspects' preference orderings is:

  *u*1 (*Fink*, *Quiet*) > *u*1 (*Quiet*, *Quiet*) > *u*1 (*Fink*, *Fink*) > *u*1 (*Quiet*, *Fink*).

- Suspect 2's ordering is (*Q*, *F*), (*Q*, *Q*), (*F*, *F*), (*F*, *Q*).

Modeling of strategic game of Prisoner's Dilemma

		Suspect 2	
		Quiet	*Fink*
Suspect 1	*Quiet*	2, 2	0, 3
	Fink	3, 0	1, 1

Each row represents
The payoff function
To the action profile of
Suspect 1, similarly
column represents for
suspect 2

Solutions of Prisoner's Dilemma

	Quiet	*Fink*
Quiet	2, 2	0, <u>3</u>
Fink	<u>3</u>, 0	<u>1</u>, <u>1</u>

Fink is the best response of each player, regardless of what the other player does. Fink is the strictly dominant and rationalizable strategy. (Fink, Fink) is the Nash Equilibrium.

Situations that can be models as Prisoner`s Dilemma (but not limited)

- *Two friends working on joint project*
- *Model ?*

	Work hard	*Goof off*
Work hard	2, 2	0, 3
Goof off	3, 0	1, 1

- *Duoploy:* two firms produce the same good, for which each firm charges either a low price or a high price. Each firm wants to achieve the highest possible profit.

- *The arms race: Two countries are racing each other to build nuclear arm.*

Bach or Stravinsky

Two daters would rather be together than separate, but dater 1 prefers Bach and dater 2 prefers Stravinsky.

- *Players:* The two daters.

- *Strategies:* Each dater's strategy set is *{Bach, Stravinsky}*.

- *Preferences:* Dater 1's ordering of the strategy profiles, from best to worst is (*B, B*), (*S, S*), (*B, S*) = (*S, B*).
- Payoff functions that represent the Dater 1's preference orderings is:

  $u1\ (B\ ,\ B):2 > u1\ (S,\ S):1 > u1\ (\mathrm{B},\ S) = u1\ (S,\ B):0.$

- Dater 2's ordering is (*S, S*), (*B, B*), (*S, B*) = (*B, S*).

  $u2\ (S\ ,\ S):2 > u2\ (B,\ B):1 > u2\ (\mathrm{S},\ B) = u2\ (B,\ S):0.$

		Dater 2	
		Bach	*Stravinsky*
Dater 1	*Bach*	2, 1	0, 0
	Stravinsky	0, 0	1, 2

If they can coordinate, either the two daters go to Bach's concert or to Stravinsky's concert.

Solutions of Bach or Stravinsky

		Person 2	
		Bach	*Stravinsky*
Person 1	*Bach*	2, 1	0, 0
	Stravinsky	0, 0	1, 2

For each player, B is the best response to B, and S is the best response to S. There are two Nash Equilibria, (B, B) and (S, S). All strategies are rationalizable, and none is dominated.

Matching Pennies

Player 1 wins if the coins are matched.

Player 2 wins if they are not matched.

- *Players:* The two players.
- *Strategies:* Each player's set of actions is *{Head, Tail}*.
- *Preferences:* Player 1's ordering of the strategy profiles, from best to worst, is *(H, H) = (T, T), (H, T) = (T, H).*

  Player 2's ordering is *(H, T) = (T, H), (H, H) = (T, T).*

		Player 2	
		Head	*Tail*
Player 1	*Head*	1, -1	-1, 1
	Tail	-1, 1	1, -1

- There is no sure way to win for either of the players.
- For player 1, H is the best response to T and viceversa, for player 2, H is the best response to H and T is the best response to T.
- All strategies are rationalizable and none is dominated. There are no Nash Equilibria.

Another Game: The Stag Hunt

The strategic game that corresponds to this specification is:

Players: The hunters.
Actions: Each player's set of actions is {*Stag*, *Hare*}.

Preferences:

- For each player, the action profile in which all players choose *Stag* (resulting in her obtaining a share of the stag) is ranked highest,
- followed by any profile in which she chooses *Hare* (resulting in her obtaining a hare),
- followed by any profile in which she chooses *Stag* and one or more of the other players chooses *Hare* (resulting in her leaving empty-handed).

Another Game: The Stag Hunt

		Player 2	
		Stag	*Hare*
Player 1	*Stag*	2, 2	0, 1
	Hare	1, 0	1, 1

Game or Nash Equilibria

- Games also often have solutions or equilibrium points.
- These are outcomes which, owing to the selection of particular reasonable strategies will result in a determined outcomes.
- An equilibrium is that point where it is not to either players advantage to unilaterally change his or her mind.

Saddle points

- The Nash equilibrium is also called a saddle point because of the two curves used to construct it:
  - an upward arching Maximin gain curve
  - and a downward arc for minimum loss.
  - Draw in 3-d, this has the general shape of a western saddle (or the shape of the universe; and if you prefer). .

Nash Equilibrium

A (pure-strategy) Nash equilibrium is a strategy profile s^* (or action profile a^*) with the property that no player i can do better by choosing a strategy different from s_i^*, given that every other player j adheres to s_j^*.

Definition The strategy profile s^* is a *Nashequilibrium* if, for every player i, $u_i(s^*) \geq u_i(s'_i, s_{-i}^*)$ for every strategy s'_i of player i.

Understanding Nash Equilibrium:

- Let a be an action profile, in which the action of each player i is a_i. Let a'_i be any action of player i (either equal to a_i, or different from it). Then (a'_i, a_{-i}) denotes the action profile in which every player j *except* i chooses her action a_j as specified by a, whereas player i chooses a'_i. (The $-i$ subscript on a stands for "except i".)

- That is, (a'_i, a_{-i}) is the action profile in which all the players other than i adhere to a while i "deviates" to a'_i. (If $a'_i = a_i$ then of course $(a'_i, a_{-i}) = (a_i, a_{-i}) = a$.)

- If there are three players, for example, then (a'_2, a_{-2}) is the action profile in which players 1 and 3 adhere to a (player 1 chooses a_1, player 3 chooses a_3) and player 2 deviates to a'_2.

Nash equilibrium of strategic game with ordinal preferences

- On the basis of above understanding, the definition of Nash-Equilibrium can be restated as:

The action profile a^* in a strategic game with ordinal preferences is a **Nash equilibrium** if, for every player i and every action a_i of player i, a^* is at least as good according to player i's preferences as the action profile (a_i, a^*_{-i}) in which player i chooses a_i while every other player j chooses a^*_j . Equivalently, for every player i,

$$u_i(a^*) \geq u_i(a_i, a^*_{-i}) \text{ for every action } a_i \text{ of player } i,$$

where u_i is a payoff function that represents player i's preferences.

Nash equilibrium of strategic game with ordinal preferences

- The definition of a Nash equilibrium is designed to model a steady state among experienced players.

- An alternative approach to understanding players' actions in strategic games assumes that the players know each others' preferences, and considers what each player can deduce about the other players' actions from their rationality and their knowledge of each other's rationality.

There are two main justifications for Nash Equilibrium:

- Self-Enforcing Contract.

  The players meet and agree before playing on the course of actions s^*. The contract s^* is self-enforcing if no player has reasons to deviate if the others do not.

- Learning Equilibrium Play.

  The play s^* is in equilibrium if no player i would deviate, were she to learn the opponents' play s_{-i}^*, because of communication, observation, or repetition.

Nash Equilibrium of previous discussed games

1. Prisoner's Dilemma & Exercise 25.1 and 25.2 in chapter 2,pp 25
2. BoS
3. Matching pennies
4. Stag Hunt (Hawk-Dove (exercise 29.1))

Discuss the Nash-equilibria of each game.

Dominance

A player's strategy strictly dominates another one if it gives a higher payoff, no matter of what other players do.

Definition Player i's strategy s_i *strictly dominates* strategy s'_i if $u_i(s_i, s_{-i}) > u_i(s'_i, s_{-i})$ for every profile s_{-i} of opponents' strategies.

Theorem *A strictly dominated strategy s_i is never part of any Nash equilibrium s^*.*

2.9.1 in Chapter 2 by Osborn

A player's strategy weakly dominates another strategy if it is always at least as good, and sometimes better.

Definition Player i's strategy s_i *weakly dominates* her strategy s'_i if

$u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i})$ for every profile s_{-i} of opponents' strategies

and

$u_i(s_i, s_{-i}) > u_i(s'_i, s_{-i})$ for some profile s_{-i} of opponents' strategies.

Note *There exist games with Nash Equilibria s^* that include weakly dominated strategies s_i for some player i.*

		Player 2	
		C	*D*
Player 1	*A*	$\underline{1}, \underline{1}$	$\underline{0}, 0$
	B	$0, \underline{0}$	$\underline{0}, \underline{0}$

There are two Nash Equilibria: (A,C) and (B,D).

The Nash Equilibrium (B,D) is weakly dominated.

Best Response Correspondences/Function

The best response correspondence or function B_i **,** of player *i* assigns to each profile s_{-i} of opponents' strategies, the set of player *i* 's strategies that maximizes her payoff.

Definition The *best response correspondence* B_i of player *i* is:

$B_i(s_{-i}) = \{s_i \text{ in } S_i : u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i}) \text{ for all } s'_i \text{ in } S_i\}$.

- Any action in $B_i(a_{-i})$ is at least as good for player *i* as every other action of player *i* when the other players' actions are given by a_{-i}. We call $\boldsymbol{B_i}$ the **best response function** of player *i*.

Proposition *The strategy profile s^* is a Nash equilibrium of a game G=(I, S, u) if and only if every player's strategy is a best response to the other players' strategies: s^*_i belongs to $B_i(s^*_{-i})$ for every player i*.

Lecture 2

Models of Oligopoly: Cournot, Bertrand, and Stackelberg

Cournot: Quantity Setters

Bertrand: Price setters

Stackelberg: First Mover Advantages

Illustration: Cournot`s Model of Oligopoly: Quantity Setters

KEY TAKEAWAYS

- Cournot competition is an economic model in which competing firms choose a quantity to produce independently and simultaneously.
- The model applies when firms produce identical or standardized goods.
- It's assumed that they can't collude or form a cartel.
- The concept that one firm will react to what it believes a rival will produce forms part of the perfect competition theory.

Illustration: Cournot`s Model of Oligopoly

- Model used for understanding the interaction among smaller number of firms.
- Two models are discussed in 18th centaury before Nash Notion.

General Model

- A single good is produced by n firms.

- Firm i's cost of producing q_i units is $C_i(q_i)$.
  C_i is an increasing function.

- All the output is sold at a single price, determined by the demand for the good and the firms' total output. The firms' total output is

$$Q = q_1 + \ldots + q_n.$$

- The market price is P(Q).
  P is the inverse demand function, decreasing if positive.

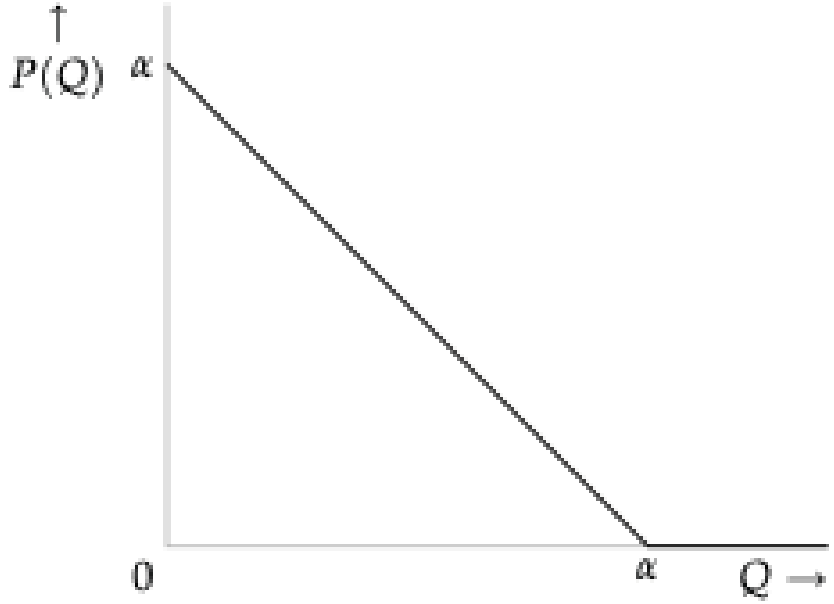


Fig: The inverse demand function in the duopoly of Cournot's game

Illustration: Cournot`s Model of Oligopoly

- Cournot suggested that the industry be modeled as the following strategic game, which is refered to as **Cournot's oligopoly game**.

Players: The firms.

Actions: Each firm's set of actions is the set of its possible outputs (nonnegative numbers).

Preferences: Each firm's preferences are represented by its profit

Linear Costs and Demand

- If the output of each firm i is q_i, then the market price is

$$P(q_1 + \cdots + q_n),$$

then Firm i's revenue is $q_i P(q_1 + \ldots + q_n)$.

- Firm i's profit is revenue minus its cost:

$$\pi_i(q_1 + \ldots + q_n) = q_i P(q_1 + \ldots + q_n) - C_i(q_i).$$

- <u>Example: duopoly with constant unit cost and linear inverse demand function</u>
- Suppose there are two firms
- each firm's cost function is the same, given by $C_i(q_i) = cq_i$ for all q_i
- $P(Q) = \alpha - Q$ if $\alpha > Q$, $P(Q) = 0$ if $\alpha < Q$. (alpha maximum price the market is willing to pay when the total quantity Q = 0)
- $\pi_i(q_1, \ldots, q_n) = q_i [\alpha - (q_1 + \ldots + q_n)] - cq_i$. ---------- Eq. (1)
- So if there are two firms and If the firms' outputs are $q1$ and $q2$ then the market price $P(q1 + q2)$ is $\alpha - (q1 + q2)$ if $q1 + q2 \leq \alpha$ and zero if $q1 + q2 > \alpha$. Thus firm 1's profit is

$$\begin{aligned} \pi_1(q_1, q_2) &= q_1(P(q_1 + q_2) - c) \\ &= \begin{cases} q_1(\alpha - c - q_1 - q_2) & \text{if } q_1 + q_2 \leq \alpha \\ -cq_1 & \text{if } q_1 + q_2 > \alpha. \end{cases} \end{aligned}$$

- To find the Best Response Functions, differentiate π_i with respect to q_i in equation(1), and set it equal to zero, and obtain:

  $d\pi_i (q_1, \ldots, q_n)/dq_i = \alpha - q_i - (q_1 + \ldots + q_n) - c = 0.$

- Best Response functions:

  $b_i (q_i) = [\alpha - (q_1 + \ldots + q_{i-1} + q_{i+1} + \ldots + q_n) - c]/2.$

- To find the Nash equilibria, we solve the system of best-response functions.

- Because this system is linear and symmetric, we equalize q_i^* across i = 1,…,n:

  $$q_i^* = b_i (q_i^*) = [\alpha - (n-1)\, q_i^* - c]/2.$$

- Solving the above equation, we find that the Nash equilibrium quantity is:

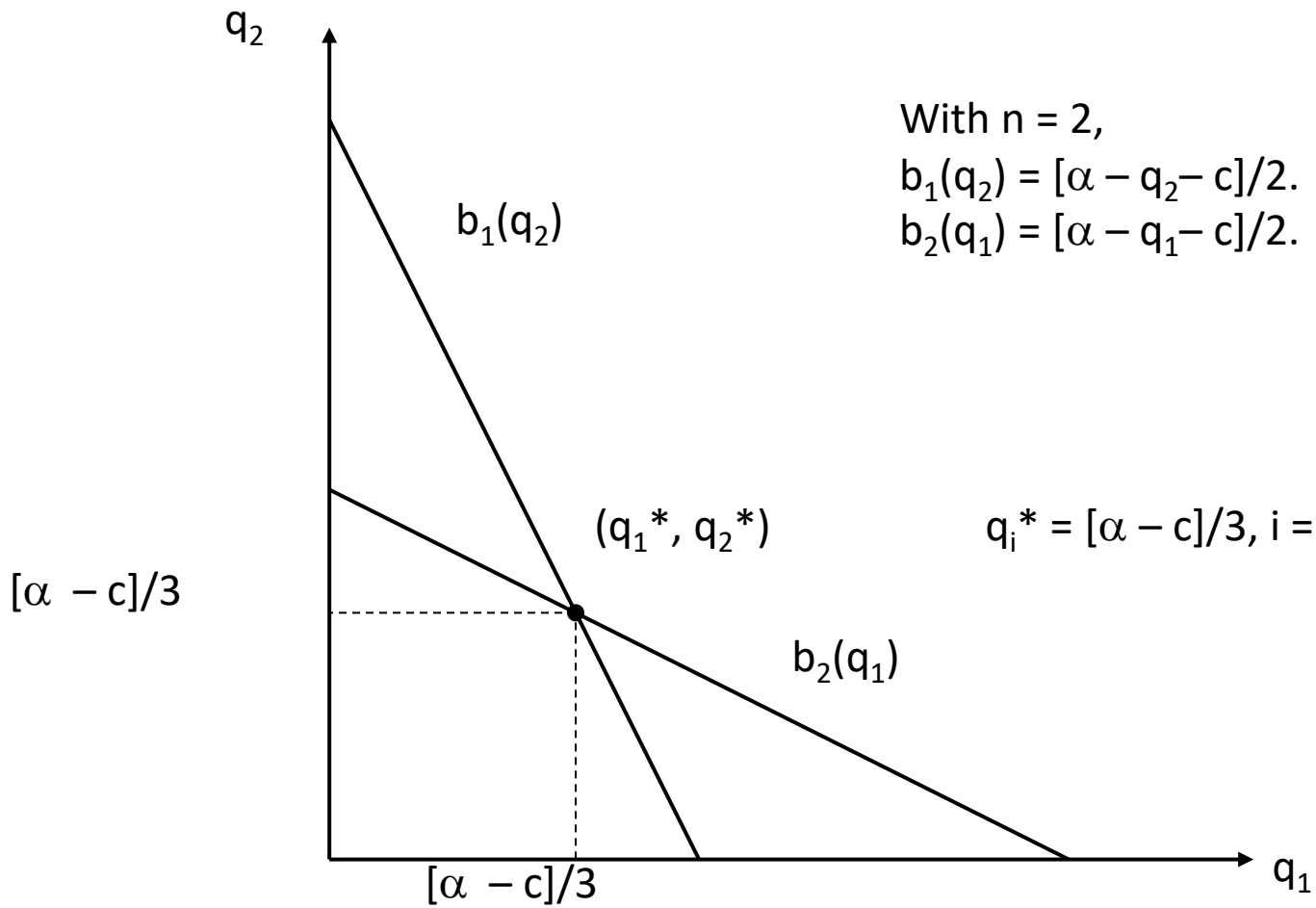
$$q_i^* = [\alpha - c]/(n+1).$$

- Substituting in the formula for the price, we find that the Nash equilibrium price is:

$$p_i(q_i^*) = \alpha - Q^* = \alpha - n[\alpha - c]/(n+1)$$
$$= [\alpha + nc]/(n+1).$$

- The Nash equilibrium profits are:

$$\pi_i(q_i^*) = q_i^*[\alpha - Q^*] - cq_i^*$$
$$= [\alpha - c]^2/(n-1)^2.$$



With n = 2,

$b_1(q_2) = [\alpha - q_2 - c]/2.$

$b_2(q_1) = [\alpha - q_1 - c]/2.$

$q_i^* = [\alpha - c]/3, i = 1,2.$

Bertrand Competition: Price setters

- Unlike Cournot competition, firms compete in prices.
- The demand function is denoted by D, if the good is available at the price p, then the total amount demanded is D(p).
- The firm setting the lowest price sells to all the market.

Linear Costs and Demand

- $C_i(q_i) = cq_i$, i=1, …, n.

$D(p) = \alpha - p$ if $\alpha > p$,
$D(p) = 0$ if $\alpha < p$.

- Let $\underline{p}_i = \min \{p_j, j \text{ different from } i\}$. Here, p_j: price of rest firms

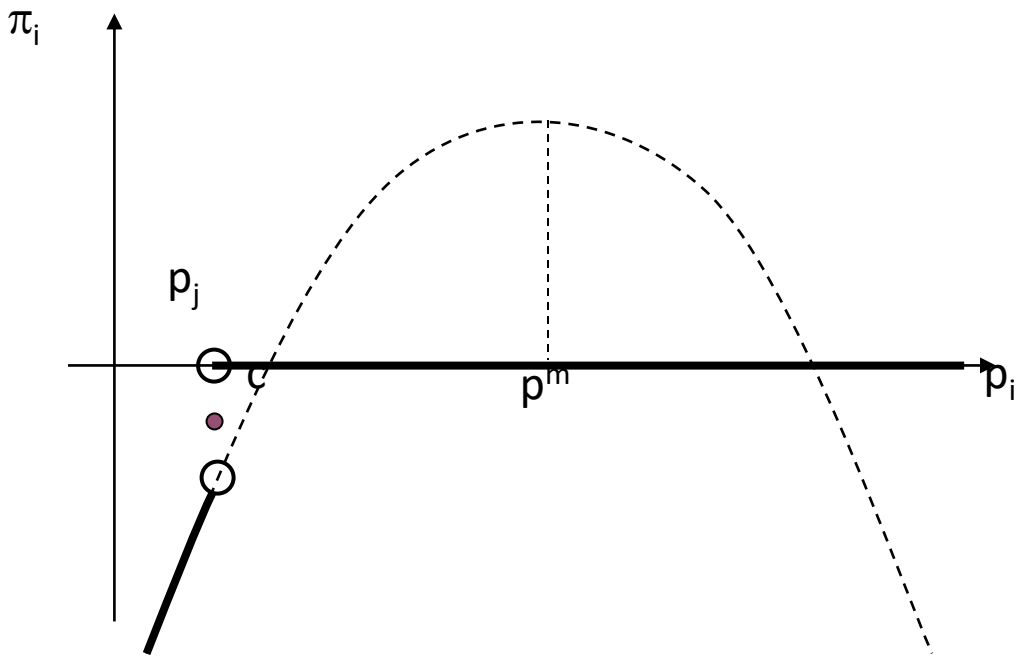
- The profit is:

$\pi_i(p_1, \ldots, p_n) = (p_i - c)(\alpha - p_i)$ if $p_i < \underline{p}_i$,
$\pi_i(\mathbf{p}) = (p_i - c)(\alpha - p_i)/|\{k : p_k = \underline{p}_i\}|$ if $p_i = \underline{p}_i$,
$\pi_i(\mathbf{p}) = 0$ if $p_i > \underline{p}_i$.

Best-Response Correspondence

Suppose that there only two firms, so that $\underline{p}_i = p_j$.

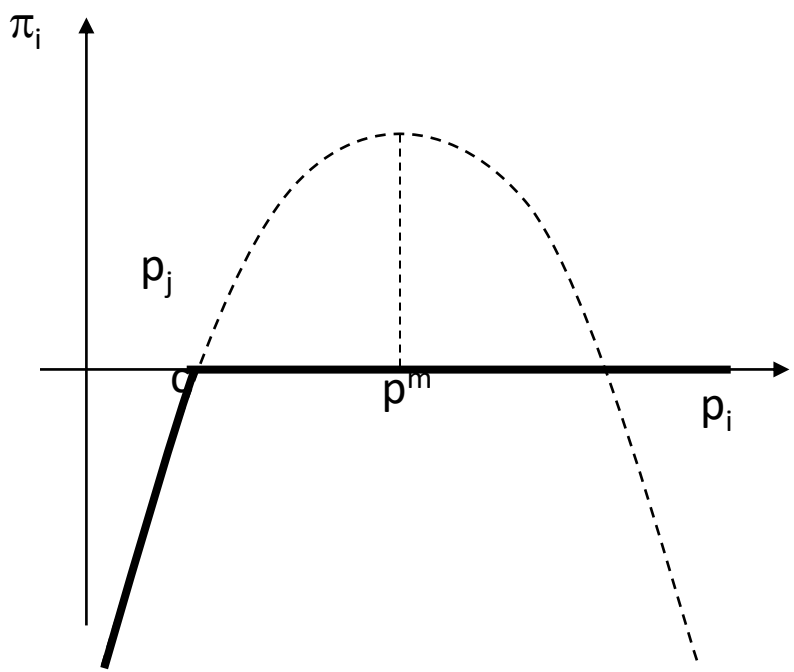
- Case 1: If $p_j < c$, then $\pi_i(\mathbf{p}) < 0$ for $p_i \leq p_j$, {here $p_i = p_j < c$}
  $\pi_i(\mathbf{p}) = 0$ for $p_i > p_j$: $b_i(\mathbf{p}) = \{p_i : p_i > p_j\}$.

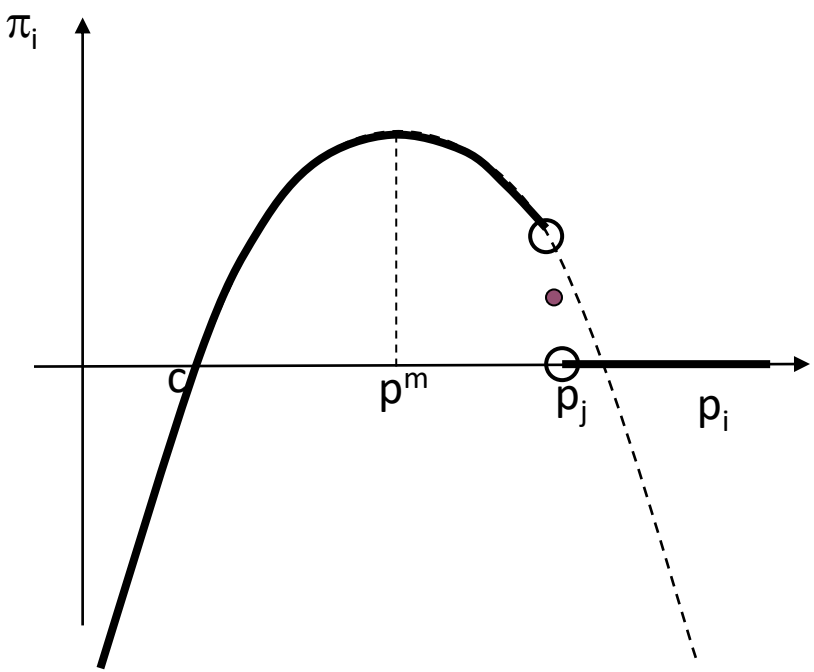


Case 2: If $p_j = c$, then $\pi_i(\mathbf{p}) < 0$ for $p_i < p_j$,

$\pi_i(\mathbf{p}) = 0$ for $p_i \geq p_j$: $b_i(\mathbf{p}) = \{p_i : p_i \geq p_j\}$.

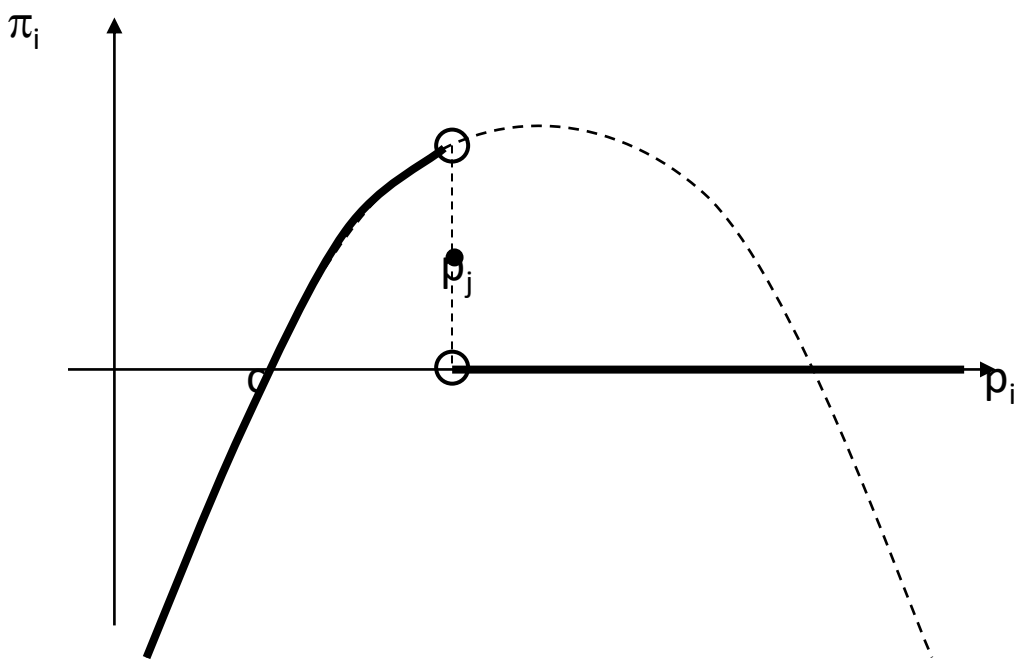
Case 3: If $p_j > p^m$, then $b_i(\mathbf{p}) = \{p^m\}$.





- If $c < p_j \leq p^m$ then $\pi_i(\mathbf{p})$ increases in p_j, but discontinuously drops at $p_i = p_j$. So, $b_i(\mathbf{p}) = \phi$.

  The best response correspondence is empty.



In sum, the best-response correspondence is:

- $b_i(\mathbf{p}) = \{p_i : p_i > p_j\}$, if $p_j < c$,
- $b_i(\mathbf{p}) = \{p_i : p_i \geq p_j\}$, if $p_j = c$,
- $b_i(\mathbf{p}) = \phi$, if $c < p_j \leq p^m$,
- $b_i(\mathbf{p}) = \{p^m\}$, if $p_j > p^m$.

The Nash equilibrium is $p_i = c$, for all $i = 1,\ldots,n$.

Intuitively, selling at any price $p_i < c$ yields negative profit. If the lowest industry price were $p_j > c$, then firm i sells to the whole industry at any price p_i with $c < p_i < p_j$. In equilibrium, $p_i = c$, for all i.

Downsian Electoral Competition: Based on Hotelling Model

- Harold Hoteling analysed a model of spatial competition; i.e. the location of different sellers in a market respect to one another.
- how sellers would choose locations along a a linear market. E.g: **ice cream stands along a beach**

Analogy:

1. The product is uniform so customers would buy from the nearest seller.
2. Suppose that there are only two sellers and they are located at the quarter points of the beach. The two sellers would then each capture one half of the beach ice cream market.

Case 1: <u>If the customers were evenly distributed along the beach</u>:

Case 2: <u>If the customers were not evenly distributed along the beach</u>

Downsian Electoral Competition:Continued

Case 1: <u>If the customers were evenly distributed along the beach</u>:

- The arrangement would have the advantage that it would reduce the amount of walking by the customers to a minimum. But that arrangement is not stable.
- Seller M would have a profit incentive to move toward Seller N, in fact to move just to the west of Seller N and thereby capture three quarters of the ice cream market.
- Likewise Seller N would have an incentive to move toward Seller M.
- They might chose the locational advantage but the only place they could settle down is at the midpoint of the beach.
- Again each gets one half of the market, but now the average distance customers have to walk is double what it was when the sellers were located at the quarter points.

Downsian Electoral Competition:Continued

Case 2: <u>If the customers were not evenly distributed along the beach</u>

- If the customers were not evenly distributed along the beach the tendency of the ice cream sellers under competition would be to locate at the median point, the point where half of the customers are to the west of that point and half are to the east of it.

- Anthony Downs noted that Hoteling's model could explain political competition.
- Instead of a beach the linear market is the political spectrum.
- The assumption is that each voter will vote for the candidate or party that is closest to his or her political position. Thus when a candidate takes a position to the right of the other candidate he or she will get the votes of all to the right of that position.
- As with the ice cream sellers on the beach the political candidates will choose a political position that is virtually the same as their opponent's. Furthermore the candidates will be driven to select the political position of the median voter.

Downsian Electoral Competition: Based on Hotelling Model

- *The players* are 2 candidates in an election.

- *A strategy* is a real number x, representing a policy(known as position) on the left-right political ideology spectrum.

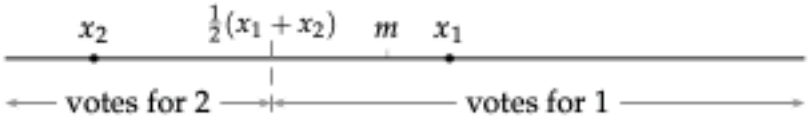


Figure 70.2 An action profile (x_1, x_2) for which candidate 1 wins.

- After the candidates choose policies, each citizen votes for the candidate with the policy she prefers.

- The candidate who obtains the most votes wins. Candidates care only about winning.

Figure is taken from M. Osborn page no:71

- The voters are a continuum with diverse ideologies or favorite position , with cumulative distribution F.

- For any k, a voter with ideology or favorite position y (x* in reference book), is indifferent between the policies y - k and y + k.

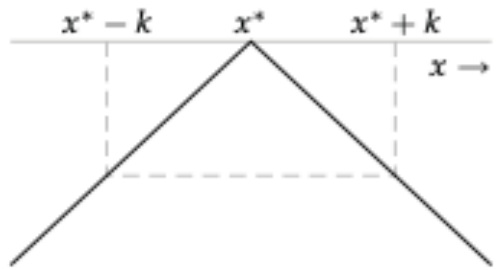


Figure 69.1 The payoff of a voter whose favorite position is x^*, as a function of the winning position, x.

- The median m is such that 1/2 of voters' has ideologies y $\geq$ m, and 1/2 has ideologies y $\leq$ m. So, F (m) = 1/2.

- each candidate attracts the votes of all citizens whose favorite positions are closer to her position than to the position of any other candidate.
- Consider there are three candidates, with positions x1, x2, and x3 as shown:

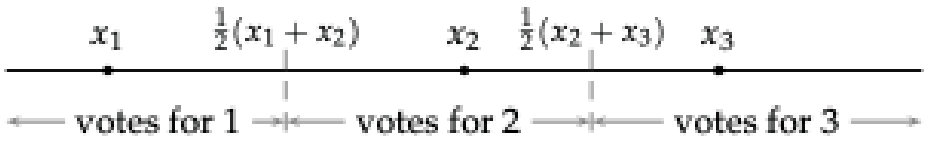


Figure 70.1 The allocation of votes between three candidates, with positions x_1, x_2, and x_3.

- Candidate 1 attracts the votes of every citizen whose favorite position is in the interval, labeled “votes for 1”, up to the midpoint 1/2 (x1 + x2) of the line segment from x1 to x2.
- candidate 2 attracts the votes of every citizen whose favorite position is in the interval from 1/2 (x1 + x2) to 1 (x2 + x3)
- and candidate 3 attracts the remaining votes.

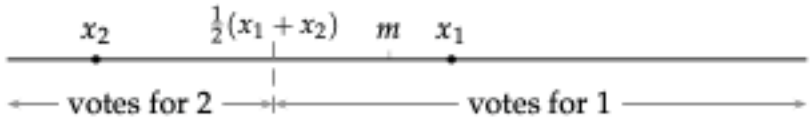


Figure 70.2 An action profile (x_1, x_2) for which candidate 1 wins.

- Fix the position x2 of candidate 2 and consider the best position for candidate 1.

<u>First suppose that x2 < m.</u>

- If candidate 1 takes a position to the left of x2 then candidate 2 attracts the votes of all citizens whose favorite positions are to the right of 1/2 (x1 + x2), a set that includes the 50% of citizens whose favorite positions are to the right of m, and more. Thus candidate 2 wins, and candidate 1 loses.
- If candidate 1 takes a position to the right of x2 then she wins so long as the dividing line between her supporters and those of candidate 2 is less than m.
- She prefers to win than to lose, and is indifferent between all the outcomes in which she wins, so her set of best responses to x2 is the set of positions that causes the midpoint ½(x1+x2) of the line segment from x2 to x1 less than m.
- If the mid point is equal to m, then she losses.
- The condition ½(x1+x2)<m => x1 < 2m-x2, so the candidate 1 best response to x2 is set of all position between x2 and 2m-x2.

Best Response Functions

- Fix the policy x_2 of candidate 2 and consider 1's choice.
- Suppose that $x_2 < m$, the case for $x_2 > m$ is symmetric.
- If candidate 1 chooses $x_1 < x_2$ then she wins the votes of citizens with ideology $y < ½ (x_1 + x_2)$.
- Because $½ (x_1 + x_2) < x_2 <$ follows that

  $F(½ (x_1 + x_2)) < ½$, so that can 1 wins less than

  ½ of the votes, and loses the election.

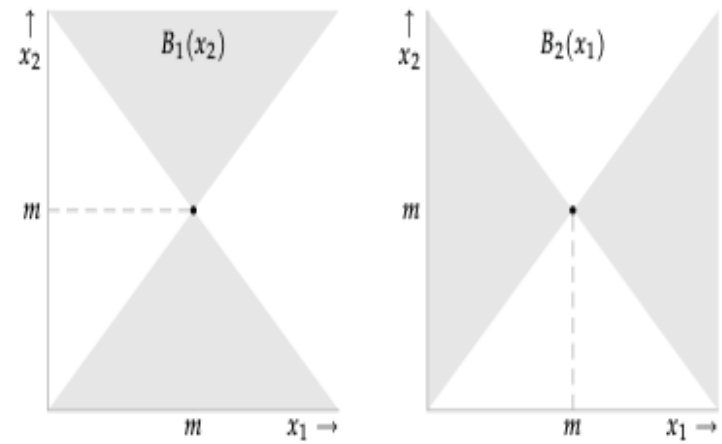


Figure 71.1 The candidates' best response functions in Hotelling's model of electoral competition with two candidates. Candidate 1's best response function is in the left panel; candidate 2's is in the right panel. (The edges of the shaded areas are excluded.)

$$B_1(x_2) = \begin{cases} \{x_1 : x_2 < x_1 < 2m - x_2\} & \text{if } x_2 < m \\ \{m\} & \text{if } x_2 = m \\ \{x_1 : 2m - x_2 < x_1 < x_2\} & \text{if } x_2 > m. \end{cases}$$

- If $x_1 > x_2$, then candidate 1 wins the votes of citizens with ideology $y > ½ (x_1 + x_2)$.

  She wins more or less than ½ of the votes if and only if

  $1 – F(½ (x_1 + x_2)) > ½$.

  In this case, she wins the election.

- This is equivalent to $½ (x_1 + x_2) < m$, i.e. $x_1 < 2m - x_2$.

- So, $b_1 (x_2) = \{x_1 : x_2 < x_1 < 2m - x_2 \}$ for $x_2 < m$.

- For $x_2 > m$, $b_1(x_2) = \{x_1 : 2m - x_2 < x_1 < x_2\}$.

- If $x_2 = m$, then player 1 loses the election unless she plays $x_1 = m$. So $b_1(m) = \{m\}$.

- By using the best response correspondences the unique Nash Equilibrium is (m, m). The candidates' political platforms converge to the median policy.

- Intuitively, consider any pair of platforms (x_1, x_2) other than (m, m). One candidate can win the election by deviating and locating ε closer to m than x_2.

  Hence (x_1, x_2) is not a Nash Equilibrium.

- If instead x_1=m, then candidate 2 loses the election for sure unless she plays x_2 = m.

  Hence (m, m) is a Nash Equilibrium.

Vickrey Second-Price Sealed-bid Auctions

- In an “English” auction, n bidders submit increasing bids for a good, until only one is left, who wins the auction.

- The price paid by the last bidder is her last bid.

- Suppose each bidder’s valuation of the good is independent of the other bidders’ values.

  For example, Vickrey’s model applies when the good is a work of art, but not when it is a oil field.

- The English auction is equivalent to a sealed-bid auction, in which each bidder decides, before bidding begins, the most she is willing to bid.

- To win, the bidder with the highest valuation needs to bid slightly more than the second highest maximal bid.

- If the bidding increment is small, the price the winner pays equals the second highest maximal bid.

Second-Price Auction Game

- Players: n bidders. Bidder i's valuation is v_i(*maximal-bid*), we order $v_1 > \ldots > v_n > 0$, without loss of generality.

- Strategies: bidder i's maximal sealed-bid is b_i.

- Let $\underline{b}_i$ (*second highest bid*)= max $\{b_j : j \text{ different from } i\}$.

- Payoffs: $u_i(b_1, \ldots, b_n) = \begin{cases} v_i - \underline{b}_i & \text{if } b_i > \underline{b}_i \\ 0 & \text{if } b_i < \underline{b}_i \end{cases}$

Second-Price Auction Game as Strategic game

In summary, a **second-price sealed-bid auction** (with perfect information) is the following strategic game.

- *Players:* The n bidders, where $n \geq 2$.

- *Actions:* The set of actions of each player is the set of possible bids (nonnegative numbers).

- *Preferences:* The payoff of any player i is $v_i - \underline{b}_i$, where $\underline{b}_i$ is the highest bid submitted by a player other than i if either b_i is higher than every other bid, or b_i is at least as high as every other bid and the number of every other player who bids b_i is greater than i. Otherwise player i's payoff is 0.

Nash Equilibria

This game has many Nash equilibria.

1. $(b^*_1, \ldots, b^*_n) = (v_1, \ldots, v_n)$.

Since $v_1 > v_2 > \cdots > v_n$, the outcome is that player 1 obtains the object at the price b2

Bidder 1 wins the object, payoff: $v_1 - b^*_2 = v_1 - v_2 > 0$.
If bidding $b_1 < v_2$, she loses the object, the payoff is 0.
If bidding $b_1 > v_2$, her payoff is $v_1 - v_2 > 0$.

The payoff of bidders i = 2, …, n is 0. (remaining bidders)

If bidding $b_i > v_1$, the payoff is $v_i - b_1 = v_i - v_1 < 0$.
If bidding $b_i < v_1$, she loses the object, the payoff is 0.

2. $(b^*_1, \ldots, b^*_n) = (v_1, 0, \ldots, 0)$

Bidder 1 wins the object, her payoff is v_1.

The payoff of bidders $i = 2, \ldots, n$ is 0.

If bidding $b_i > v_1$, the payoff is $v_i - b_1 = v_i - v_1 < 0$.

If bidding $b_i < v_1$, she loses the object, the payoff is 0.

3. $(b^*_1, \ldots, b^*_n) = (v_2, v_1, 0, \ldots, 0)$

Bidder 2 wins the object, payoff $v_2 - b_1 = v_2 - v_2 = 0$.

To win, bidder $i = 1, 3, \ldots, n$ must bid $b_i > b_2 = v_1$,

so the payoff is $v_i - b_1 < v_i - v_1 \leq 0$.

Any of these bidders' payoff is at least as good if losing the good.

Weakly Dominant Solution

Exercise 83.1: Second-price sealed-bid auction with two bidders) Find *all* the Nash equilibria of a second-price sealed-bid auction with two bidders. (Construct the players' best response functions.

The Nash Equilibrium $(b^*_1, \ldots, b^*_n) = (v_1, \ldots, v_n)$ is the unique weakly dominant solution.

	$\underline{b}_i < b_i$ or $\underline{b}_i = b_i$ & i wins	$b_i < \underline{b}_i < v_i$ or $\underline{b}_i = b_i$ & i loses	$\underline{b}_i > v_i$
$b_i < v_i$	$v_i - \underline{b}_i$	0	0
$b_i = v_i$	$v_i - \underline{b}_i$	$v_i - \underline{b}_i$	0

	$\underline{b}_i \leq v_i$	$v_i < \underline{b}_i < b_i$ or $\underline{b}_i = b_i$ & i wins	$\underline{b}_i > b_i$ or $\underline{b}_i = b_i$ & i loses
$b_i = v_i$	$v_i - \underline{b}_i$	0	0
$b_i > v_i$	$v_i - \underline{b}_i$	$v_i - \underline{b}_i$ (< 0)	0

In sum, bidding $b_i = v_i$ yields at least as high a payoff as bidding $b_i > v_i$ or $b_i < v_i$ for any opponents' bids.

Summary of the Lecture

- Definition of Games in Strategic Form.

- Definition of Solution Concepts
  Nash Equilibrium, Dominance and Rationalizability.

- Applications
  Cournot Oligopoly, Bertrand Duopoly, Downsian Electoral Competition, Vickrey Second Price Auction.

Topics in next lecture

- Mixed Strategies.
- Stackelberg Duopoly.